CHAPTER -1

MAJOR DISEASES OF TOMATO

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INTRODUCTION

Tomato (*Solanum lycopersicum*) is a key horticultural crop, cultivated worldwide for its culinary importance and nutritional benefits. As a staple in diets and a valuable agricultural commodity, tomato production plays a critical role in both subsistence farming and commercial agriculture. However, tomato cultivation is heavily impacted by a variety of diseases, which can drastically reduce yields, compromise fruit quality, and cause significant economic losses.

These diseases, caused by a range of pathogens including fungi, bacteria, viruses, and nematodes, as well as environmental stresses, thrive under different climatic conditions, making tomato vulnerable across regions. With the increasing demand for high-quality, disease-free produce, managing tomato diseases has become a primary concern for growers and researchers alike.

In this chapter, we will examine the most prevalent diseases affecting tomato plants, their symptoms, modes of infection, and the conditions that promote their spread. We will also explore modern strategies for disease prevention and control, including the use of resistant varieties, cultural practices, and integrated disease management techniques, all aimed at ensuring sustainable tomato production and minimizing the adverse effects of these diseases.

MAJOR DISEASE OF TOMATO

I. Damping Off: *Pythium aphanidermatum*

1. Symptoms



Damping off of tomato seedlings occurs in two distinct phases: the pre-emergence and post-emergence phases.

- **Pre-emergence phase:** Seedlings are killed before they break through the soil surface. This is characterized by the rotting of the young radicle and plumule, leading to complete seedling failure.
- **Post-emergence phase:** After seedlings emerge, the juvenile tissues around the collar at the soil level become infected. These tissues soften, becoming water-soaked, and the seedlings collapse or topple over.

2. Mode of Spread and Survival

Pythium aphanidermatum and other causal organisms responsible for damping off are soil-borne pathogens. They persist in the soil, especially in areas with hosts available for infection. These pathogens generally have a wide host range, making them difficult to control in heavily cultivated areas.

3. Management

- Use raised seed beds to improve soil aeration and prevent waterlogging.
- Provide light but frequent irrigation to ensure proper drainage.
- Drench the soil with Copper oxychloride (0.2%) or Bordeaux mixture (1%) to reduce pathogen load.
- Seed treatment with beneficial fungal cultures like *Trichoderma viride* (4 g/kg of seed) or chemical fungicides like Thiram (3 g/kg of seed) can effectively prevent pre-emergence damping off.
- In cloudy weather, spraying Metalaxyl (0.2%) is recommended to control post-emergence damping off.

II. Fusarium Wilt: *Fusarium oxysporum* f. sp. *lycopersici*

1. Symptoms

The initial signs of Fusarium wilt in tomatoes include:

- Clearing of the veinlets and chlorosis (yellowing) of the leaves.
- Younger leaves die progressively, leading to the wilting and eventual death of the plant within a few days.
- The petioles and leaves droop and wilt, especially in young plants, which show vein clearing and petiole drooping.
- In the field, the disease starts with the yellowing of lower leaves, followed by wilting and death of the leaflets.
- Symptoms continue to appear on subsequent leaves, eventually leading to the browning of the vascular system.
- In advanced stages, the plant becomes stunted and dies.



2. Pathogen

- The fungus produces septate, hyaline mycelium.
- It generates both macroconidia and microconidia.
- Microconidia are one-celled, hyaline, and vary from ovoid to ellipsoid in shape.
- Two distinct races of the pathogen have been identified, which differ in their virulence.

3. Mode of Spread and Survival

- *Fusarium oxysporum* f. sp. *lycopersici* is both seed-borne and soil-borne.
- It survives in the soil for over 10 years as chlamydospores or in infected crop debris as saprophytically growing mycelium.
- Distribution of the pathogen occurs through seedlings raised in infected soil, as well as wind-borne spores, surface drainage water, and contaminated agricultural implements.

4. Management

- Remove and destroy infected plants to prevent the spread of the disease.
- Spot drenching with Carbendazim (0.1%) is effective in managing localized infections.
- Practice crop rotation with non-host crops such as cereals to reduce pathogen build-up in the soil.

III. Early Blight: *Alternaria solani*

1. Symptoms

Early blight is a common disease of tomato that can affect the plant at any growth stage, primarily attacking the foliage. Key symptoms include:

- Small black lesions first appear on older foliage.
- The lesions enlarge, developing concentric rings in a bull's-eye pattern in the center of the spots.
- The surrounding tissue may turn yellow, and under favorable conditions (high temperature and humidity), the foliage can die off rapidly.
- Stem lesions are similar to leaf lesions and can girdle the plant if they occur near the soil line, potentially leading to plant death.
- The fungus can also infect the fruit, particularly through the calyx or stem attachment, causing large lesions with concentric rings that may cover almost the entire fruit.



2. Pathogen

- The mycelium of *Alternaria solani* is septate, branched, and light brown, darkening with age.
- Conidiophores are dark, and the conidia are beaked, muriform, and dark-colored, borne singly.

3. Mode of Spread and Survival

- The pathogen is primarily spread by wind and rain splashes.
- Under dry conditions, it survives in infected plant debris in the soil for up to three years and can also be seed-borne.

4. Management

- Removal and destruction of infected crop debris helps prevent the spread of the disease.
- Practicing crop rotation can minimize disease incidence.
- Regular spraying of the crop with Mancozeb (0.2%) is effective in controlling the disease.

IV. Septoria Leaf Spot: *Septoria lycopersici*

1. Symptoms

Septoria leaf spot can affect tomato plants at any growth stage and is characterized by:

- Numerous small, gray, circular spots on the leaves, each surrounded by a distinct dark border.
- As the disease progresses, severe defoliation can occur, reducing the plant's vigor and fruit production.



2. Pathogen

- The mycelium is septate, branched, and hyaline when young, becoming darker with age.
- Pycnidia (fruiting bodies of the fungus) are erumpent, meaning they break through the surface of the leaf.
- Pycnidiospores are filiform (thread-like), hyaline, and septate.

3. Mode of Spread and Survival

- The pathogen spreads through wind and rain splashes, as well as by insects, and even on the hands and clothing of people handling tomato plants.
- It survives from one season to the next in infested crop debris and on solanaceous weeds.
- The fungus can also be seed-borne, and contaminated seeds result in infected seedlings.

4. Management

- Remove and destroy affected plant parts to reduce pathogen sources.
- Seed treatment with Thiram or Dithane M-45 (2 g/kg seed) helps to control seed-borne infections.
- In the field, spraying Mancozeb (0.2%) is an effective control measure against the disease.

V. Bacterial Wilt: *Burkholderia solanacearum*

1. Symptoms

Bacterial wilt is a devastating disease of tomato, particularly favored by high soil moisture and warm temperatures. Key symptoms include:

- Rapid and complete wilting of fully grown plants, often without any prior signs of stress.
- Sometimes, lower leaves drop before the wilting begins.

- The pathogen is largely confined to the vascular system, but in severe cases, it invades the cortex and pith, causing a yellow-brown discoloration of the tissues.
- When infected stems are cut and placed in water, a white streak of bacterial ooze emerges from the cut ends, indicating the presence of the pathogen.



2. Pathogen

- *Burkholderia solanacearum* is a gram-negative, rod-shaped bacterium, often found in pairs and motile with 1–4 flagella.
- The optimum growth temperature is between 30–37°C.

3. Mode of Spread and Survival

- The bacterium survives in the soil and spreads through irrigation water and infected seedlings.
- It can persist for up to 3 years in fallow soil and for extended periods in cultivated soils.
- Several plants, including chili, eggplant, groundnut, potato, and tobacco, serve as alternative hosts, enabling the bacterium to survive between tomato crops.

4. Management

- Avoid mechanical damage to seedlings during transplanting to reduce infection risk.
- Apply bleaching powder (10 kg/ha) to reduce soil-borne bacterial populations.
- Crop rotations involving combinations such as cowpea-maize-cabbage, okra-cowpea-maize, maize-cowpea-maize, and finger millet-eggplant have been found effective in reducing the incidence of bacterial wilt in tomato.

VI. Bacterial Leaf Spot: *Xanthomonas campestris* pv. *vesicatoria*

1. Symptoms

Bacterial leaf spot is favored by moist weather and splattering rains, with outbreaks often linked to heavy rainstorms. Key symptoms include:

- Small, brown, water-soaked spots on infected leaves, usually circular and surrounded by a yellowish halo.
- On older plants, the infection primarily affects older leaves, sometimes causing severe defoliation.
- The most distinctive symptoms are on green fruits, where small, water-soaked spots appear, later becoming raised and enlarged to one-eighth to one-fourth inch in diameter. The centers of these lesions turn light brown, irregular, and slightly sunken, with a rough, scabby surface.



- Ripe fruits are not susceptible to the disease, but the bacterium can contaminate the surface of seeds, surviving on the seed for extended periods.
- The pathogen also survives in alternate hosts, volunteer tomato plants, and infected plant debris.

2. Pathogen

- *Xanthomonas campestris* pv. *vesicatoria* is a gram-negative, short rod-shaped bacterium with a single polar flagellum.
- The bacterium forms capsules for protection.

3. Mode of Spread and Survival

- The pathogen survives in diseased plant debris, volunteer plants, and is seed-borne.
- It enters the plant through stomata, injuries, or lenticels and spreads via rain splashes.
- Disease spreads to new areas through infected seeds and diseased transplants.

4. Management

- Use disease-free seed and seedlings to prevent the introduction of the pathogen.
- Rotate crops with non-host species to avoid exposure to infected plant residues from previous seasons.
- Seed treatment with mercuric chloride (1:1000) is recommended for controlling seed-borne infections.
- Spraying with a combination of copper and organic fungicides at 5 to 10-day intervals, or Agrimycin-100 (100 ppm) sprayed thrice at 10-day intervals, effectively controls the disease.

VII.

1. Symptoms

Tomato mosaic virus (TMV) presents a variety of symptoms, including:

- Light and dark green mottling on the leaves, often accompanied by wilting of young leaves during sunny days, especially in the early stages of infection.
- Affected leaflets are usually distorted, puckered, and smaller than normal.
- In some cases, leaflets exhibit "fern leaf" symptoms, where they become indented and curled.
- Infected plants appear stunted, pale green, and spindly.
- The virus spreads through contact, such as clothes, hands of laborers, plant debris, and tools touching infected and healthy plants.



2. Pathogen

- The virus particles are rod-shaped, non-enveloped, and usually straight.
- The thermal inactivation point of the virus is between 85-90°C.

3. Mode of Spread and Survival

- The virus is seed-borne, with up to 94% of seeds potentially containing the virus.
- Infection often occurs during transplanting and is readily transmissible.
- Many solanaceous plants are susceptible to TMV, and the virus spreads easily via human handling, cultural operations with implements, animals, or leaf contact.

4. Management

- Select disease-free seeds from healthy plants for sowing.
- Soaking seeds in a solution of Trisodium Phosphate (90 g/liter of water) a day before sowing can help reduce disease incidence. Rinse seeds thoroughly and dry them in the shade after soaking.
- In the nursery, remove and destroy infected plants carefully.
- Do not use infected seedlings for transplanting.
- Practice crop rotation with crops other than tobacco, potato, chili, capsicum, and brinjal to reduce the spread of the virus.

VIII. Leaf Curl: Tomato Leaf Curl Virus (ToLCV)

1. Symptoms

Leaf curl disease leads to severe stunting of plants, characterized by downward rolling and crinkling of leaves. Newly emerging leaves show a slight yellow coloration before curling, while older leaves become leathery and brittle. The size of nodes and internodes is significantly reduced, giving infected plants a pale, bushy appearance with more lateral branches.



2. Pathogen

The virus particles are approximately 80 nm in diameter.

3. Mode of Spread and Survival

ToLCV is not seed- or sap-transmissible; however, seeds from infected fruits may carry the virus on their surface. It is primarily transmitted by the whitefly, Bemisia tabaci, and through grafting. A single viruliferous insect can effectively transmit the virus.

4. Management

To manage ToLCV:

- Use yellow sticky traps at a density of 12 per hectare to monitor whitefly populations.
- Plant barrier crops, such as cereals, around the field.

- Remove weed hosts that may harbor the virus.
- Establish a protected nursery in a net house or greenhouse.
- Spray with Imidacloprid (0.05%) or Dimethoate (0.05%) at 15, 25, and 45 days after transplanting to control whitefly vectors.

IX. Spotted Wilt: Tomato Spotted Wilt Virus (TSWV) and Groundnut Bud Necrosis Virus

1. Symptoms

Tomato spotted wilt disease is characterized by streaking on leaves, stems, and fruits. Young leaves develop numerous small, dark, circular spots and may exhibit a bronzed appearance, eventually turning dark brown and wilting. Fruits display spots about half an inch in diameter, often with concentric circular markings. On ripe fruit, these markings appear as alternate bands of red and yellow.



2. Pathogen

The pathogen consists of isometric particles ranging from 70 to 90 nm in diameter, with a thermal inactivation point of 40°C.

3. Mode of Spread and Survival

TSWV is primarily transmitted by thrips, including *Thrips tabaci*, *Frankliniella schultzi*, and *Frankliniella occidentalis*.

4. Management

To manage spotted wilt:

- Remove and destroy affected plants.
- Eliminate alternate or collateral hosts that harbor the virus.
- Plant barrier crops such as sorghum, maize, or bajra in 5-6 rows around the tomato field prior to planting.
- Apply Imidacloprid (0.05%) or any systemic insecticide to effectively control the thrips vector.

X. Gray Mould: *Botrytis cinerea*

1. Symptoms

Gray mould, caused by *Botrytis cinerea*, is characterized by lesions that initially appear as watery spots with a light brown or tan center. These lesions quickly progress to a soft, mushy mass, often leading to the breakdown of the fruit's skin. Within a short period, a grayish mycelium and spore clusters become visible. A halo effect, marked by small white rings, can form around entry points on young green fruits, and the condition may also manifest as "ghost spots," which may be single rings or solid white areas containing dark brown specks.



2. Pathogen

The fungus features septate, branched mycelium, which is hyaline when young but darkens as it ages. Its conidiophores are branched and produce conidia at their tips, which are either continuous or one-septate, elongated, and dark.

3. Mode of Spread and Survival

Botrytis cinerea thrives under high humidity, essential for its spore production. The ideal temperature for infection ranges from 65° to 75°F (18° to 24°C), with infections possible within 5 hours of favorable conditions. Conversely, temperatures exceeding 82°F (28°C) inhibit fungal growth and spore production.

4. Management

To effectively manage gray mould, consider the following strategies:

- Fungicide application: Regular spraying with Bordeaux mixture (1.0%) or mancozeb (0.2%) helps control the disease.
- Utilization of resistant varieties: Grow resistant cultivars, such as Vetomold, particularly in areas with a history of grey mould outbreaks.
- Adoption of Eurocross varieties, including Antincold, LMRI, and Sapsford's No.1, which exhibit increased resistance to this pathogen.

XI. Bacterial Soft Rot and Hollow Stem: *Erwinia carotovora* pv. *carotovora*

1. Symptoms



Bacterial soft rot is characterized by water-soaked lesions that rapidly expand, leading to a soft, mushy decay of the infected tissues. Affected fruits and stems often exhibit a foul odor due to the breakdown of plant cells. In addition, the hollow stem syndrome manifests as internal rotting within the stem, which becomes soft and may collapse. As the disease progresses, the outer stem tissue may remain intact, leading to a hollow appearance. In severe cases, the entire plant may wilt and die.

2. Pathogen

The causative agent, *Erwinia carotovora* pv. *carotovora*, is a gram-negative bacterium that thrives in moist conditions. The bacteria are rod-shaped and can produce enzymes that degrade plant cell walls, facilitating tissue maceration.

3. Mode of Spread and Survival

The pathogen is primarily spread through:

- Water splashes during rainfall or irrigation.
- Infected plant debris remaining in the soil, which can harbor the bacteria.

- Contaminated tools and equipment that come into contact with infected tissues. *Erwinia carotovora* can survive in plant debris and soil for extended periods, contributing to the disease's persistence in the environment.

4. Management

To manage bacterial soft rot and hollow stem, implement the following strategies:

- **Cultural practices**: Ensure proper drainage and avoid overhead irrigation to reduce humidity around the plants. Practice crop rotation to minimize disease incidence.
- **Sanitation**: Remove and destroy infected plant materials promptly to prevent the spread of the bacteria.
- **Seed treatment**: Use healthy, disease-free seeds and consider treating seeds with appropriate bactericides to reduce the risk of infection.
- **Chemical control**: In cases of severe infection, apply bactericides that target *Erwinia* species, following recommended guidelines for application rates and timing.

XII. Phoma Rot: *Phoma destructive*

1. Symptoms

Phoma rot is characterized by small, black, pimple-like eruptions on the surface of affected fruits and stems. These spots are the pycnidia, or fruiting bodies, of the fungus. The presence of thesse black lesions distinguishes this rot from other types. The disease typically develops under moderate temperatures and high humidity, leading to significant fruit decay.



2. Pathogen

The causal agent, *Phoma destructive*, produces ascospores that are irregularly arranged in two series. The spores are ellipsoid with obtuse ends, hyaline, and guttulate. The pycnidia of the fungus are dark brown and can be solitary or gregarious. The conidia produced are typically biguttulate, straight, and irregular in shape.

3. Mode of Spread

The pathogen is primarily seed-borne, meaning that infected seeds can carry the disease into new crops, making seed treatment essential for management.

4. Management

To manage Phoma rot effectively, consider the following strategies:

- **Seed treatment:** Use organomercurial fungicides to treat seeds before planting, which helps reduce the risk of infection.
- **Chemical control:** Apply zineb at a concentration of 0.2% as a foliar spray to provide adequate protection against the disease. Regular monitoring and early intervention can help mitigate the spread of the pathogen.

CONCLUSION

The cultivation of tomatoes is significantly impacted by various diseases that pose serious threats to yield and quality. This chapter has explored the major diseases affecting tomatoes, including Fusarium wilt, Early blight, Septoria leaf spot, Bacterial wilt, Bacterial leaf spot, Mosaic, Leaf curl, Spotted wilt, Gray mold, Bacterial soft rot, Hollow stem, and Phoma rot. Each of these diseases exhibits unique symptoms, pathogens, modes of spread, and management strategies that growers must understand to safeguard their crops.

Effective management practices are crucial for minimizing the impact of these diseases. Integrated approaches, including the use of disease-resistant varieties, proper sanitation, crop rotation, and timely application of fungicides and bactericides, can significantly reduce disease incidence and promote healthier plants. Moreover, monitoring environmental conditions and implementing preventive measures, such as the use of yellow sticky traps for insect vectors, can further enhance disease control efforts.

As the challenges posed by tomato diseases continue to evolve, ongoing research and development of innovative management strategies remain essential. By staying informed about the latest advancements in disease management and adopting sustainable practices, tomato growers can enhance productivity, reduce economic losses, and contribute to food security. Ultimately, a proactive approach to disease management will ensure the continued success of tomato cultivation in the face of growing threats from pathogens.